

Working Title: A Study of Low-Light Enhancement in the HVI Color Space

Anonymous Authors¹

Abstract

HVI-CIDNet (Yan et al., 2025) enhances low-light images by mapping them into the HVI (Horizontal/Vertical/Intensity) color space, processing intensity and chroma along separate branches, and supervising the output in both RGB and HVI. This thesis studies a question about that design; the question, the claims, and the experiments are fixed in the project plan before any run starts, and every reported number is read from a result file by a verification script rather than transcribed.

1. Introduction

Low-light image enhancement recovers a well-exposed image from an underexposed capture. Learned enhancers now dominate the paired benchmarks (Wei et al., 2018; Cai et al., 2023; Guo et al., 2020), and a recent line of work argues that the color space in which the network operates and is supervised matters as much as the architecture. HVI-CIDNet (Yan et al., 2025) maps images into a polar Horizontal/Vertical/Intensity (HVI) space, enhances intensity and chroma along separate interacting branches, and is trained with a loss summed over both the RGB output and its HVI transform.

This paper asks one question about that design. The question, the claims it supports, and the experiments that decide it are stated in Section 2; the results follow in Section 3.

2. Method and Protocol

Model. We use the public HVI-CIDNet implementation (Yan et al., 2025). The network takes an sRGB image, converts it to HVI, processes the intensity map and the chroma map along two interacting branches, and converts the result back to sRGB.

Protocol. All runs share one frozen protocol: the same dataset split, seeds, optimizer, schedule, and hardware (a

single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32). Checkpoints are selected on the validation split only; the test split is read once, by a separate final-evaluation script, per reported configuration. Each run records the SHA-256 of its training script, the upstream commit, and the split file.

3. Results

Results are reported here once the studies defined in the project plan have run. Every table in this section is generated from a result file, and every number is bound to the key it was read from.

4. Conclusion

We study one question about the HVI-CIDNet design under a single frozen protocol, and report every finding against its seed spread. The protocol and scripts are released with the thesis.

References

- Ai, Y., Chen, Z., Cai, Y., Zhang, Y., and Yang, X. BC-IHV: Conditioning the color space for stable rectified-flow low-light enhancement. *arXiv preprint arXiv:2608.21847*, 2026.
- Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., Sepahvand, N. M., Raff, E., Madan, K., Voleti, V., Kahou, S. E., Michalski, V., Arbel, T., Pal, C., Varoquaux, G., and Vincent, P. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, 2021.
- Brateanu, A., Balmez, R., Avram, A., Orhei, C., and Ancuti, C. LYT-Net: Lightweight YUV transformer-based network for low-light image enhancement. *IEEE Signal Processing Letters*, 32:2065–2069, 2025.
- Cai, J., Gu, S., and Zhang, L. Learning a deep single image contrast enhancer from multi-exposure images. *IEEE Transactions on Image Processing*, 27(4):2049–2062, 2018.
- Cai, Y., Bian, H., Lin, J., Wang, H., Timofte, R., and Zhang,

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

- Y. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 12504–12513, 2023.
- Chen, C., Chen, Q., Xu, J., and Koltun, V. Learning to see in the dark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3291–3300, 2018.
- Cheng, Z., Zhang, F., Yin, X., Yin, B., and Wang, H. VCR: Variance-driven channel recalibration for robust low-light enhancement. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(9):13598–13611, 2026.
- Chobola, T., Liu, Y., Zhang, H., Schnabel, J. A., and Peng, T. Fast context-based low-light image enhancement via neural implicit representations. In *Computer Vision – ECCV 2024*, pp. 413–430. Springer, 2024.
- Du, T., Li, M., Wang, H., and Guo, X. Anchor then polish for low-light enhancement. *arXiv preprint arXiv:2603.15472*, 2026.
- Feng, Y., Hou, S., Lin, H., Zhu, Y., Wu, P., Dong, W., Sun, J., Yan, Q., and Zhang, Y. DiffLight: Integrating content and detail for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 6143–6152, 2024.
- Guan, Q., Chen, X., Jin, G., Jin, J., Fan, S., Song, T., and Pan, J. Rethinking nighttime image deraining via learnable color space transformation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pp. 3565–3601, 2025.
- Guo, C., Li, C., Guo, J., Loy, C. C., Hou, J., Kwong, S., and Cong, R. Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1780–1789, 2020.
- Guo, X. and Hu, Q. Low-light image enhancement via breaking down the darkness. *International Journal of Computer Vision*, 131(1):48–66, 2023.
- Hou, J., Zhu, Z., Hou, J., Liu, H., Zeng, H., and Yuan, H. Global structure-aware diffusion process for low-light image enhancement. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pp. 79734–79747, 2023.
- Jiang, Y., Gong, X., Liu, D., Cheng, Y., Fang, C., Shen, X., Yang, J., Zhou, P., and Wang, Z. EnlightenGAN: Deep light enhancement without paired supervision. *IEEE Transactions on Image Processing*, 30:2340–2349, 2021.
- Johnson, J., Alahi, A., and Fei-Fei, L. Perceptual losses for real-time style transfer and super-resolution. In *Computer Vision – ECCV 2016*, pp. 694–711. Springer, 2016.
- Land, E. H. The retinex theory of color vision. *Scientific American*, 237(6):108–128, 1977.
- Li, C., Anwar, S., Hou, J., Cong, R., Guo, C., and Ren, W. Underwater image enhancement via medium transmission-guided multi-color space embedding. *IEEE Transactions on Image Processing*, 30:4985–5000, 2021.
- Li, M., Du, T., Wang, H., Hu, Q., and Guo, X. On the global photometric alignment for low-level vision. *arXiv preprint arXiv:2604.08172*, 2026.
- Liao, J., Hao, S., Hong, R., and Wang, M. GT-Mean loss: A simple yet effective solution for brightness mismatch in low-light image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6112–6121, 2025.
- Lin, Y., Lin, Z., Chen, H., Pan, P., Li, C., Chen, S., Wen, K., Jin, Y., Li, W., and Ding, X. JarvisIR: Elevating autonomous driving perception with intelligent image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22369–22380, 2025.
- Liu, R., Ma, L., Zhang, J., Fan, X., and Luo, Z. Retinex-inspired unrolling with cooperative prior architecture search for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10556–10565, 2021.
- Liu, X., Wu, Z., Vasluianu, F.-A., Yan, H., Ren, B., Zhang, Y., Gu, S., Zhang, L., Zhu, C., Timofte, R., et al. NTIRE 2025 challenge on low light image enhancement: Methods and results. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1196–1206, 2025.
- Ma, L., Ma, T., Liu, R., Fan, X., and Luo, Z. Toward fast, flexible, and robust low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5627–5636, 2022.
- Mittal, A., Moorthy, A. K., and Bovik, A. C. No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing*, 21(12):4695–4708, 2012.
- Mittal, A., Soundararajan, R., and Bovik, A. C. Making a “completely blind” image quality analyzer. *IEEE Signal Processing Letters*, 20(3):209–212, 2013.

- Musgrave, K., Belongie, S., and Lim, S.-N. A metric learning reality check. In *Computer Vision – ECCV 2020*, pp. 681–699. Springer, 2020.
- Nguyen, C. M., Chan, E. R., Bergman, A. W., and Wetstein, G. Diffusion in the dark: A diffusion model for low-light text recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 4134–4145, 2024.
- Pilligua, M., Serrano-Lozano, D., Peng, P., Baldrich, R., Brown, M. S., and Vazquez-Corral, J. Evaluating low-light image enhancement across multiple intensity levels. *arXiv preprint arXiv:2511.15496*, 2025.
- Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d’Alché Buc, F., Fox, E., and Larochelle, H. Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research*, 22(164):1–20, 2021.
- Recht, B., Roelofs, R., Schmidt, L., and Shankar, V. Do ImageNet classifiers generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97, pp. 5389–5400, 2019.
- Sharma, G., Wu, W., and Dalal, E. N. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research & Application*, 30(1):21–30, 2005.
- Shi, J.-Y., Li, M.-F., and Wu, L.-A. TPCNet: A low-light image enhancement network inspired by triple physical constraints. In *Computer Vision – ECCV 2026*, pp. 627–645. Springer, 2026a.
- Shi, K., Feng, Y., Hu, T., Cao, Y., Wu, P., Liang, Y., Zhang, Y., and Yan, Q. FusionNet: Multi-model linear fusion framework for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1019–1028, 2025.
- Shi, K., Ma, X., Feng, Y., Hu, T., Wu, P., Pang, G., and Yan, Q. HVI-CIDNet+: Beyond extreme darkness for low-light image enhancement. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(8):12126–12139, 2026b.
- Smith, A. R. Color gamut transform pairs. In *Proceedings of the 5th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pp. 12–19, 1978.
- Verma, A. K., Tripathi, S., Jakhetiya, V., Subudhi, B. N., and Jaiswal, S. Q-CIDNet: Perceptual quality aware color and intensity decoupling network for video quality enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1238–1244, 2025.
- Wang, T., Zhang, K., Shen, T., Luo, W., Stenger, B., and Lu, T. Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 2654–2662, 2023.
- Wang, X., Wang, Z., Guo, M., Chen, Q., Hao, Y., and Liu, Y. HAIMNet: A hierarchical adaptive interaction modulation network for low-light image enhancement. *IEEE Transactions on Image Processing*, 35:8180–8194, 2026.
- Wang, Y., Wan, R., Yang, W., Li, H., Chau, L.-P., and Kot, A. Low-light image enhancement with normalizing flow. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 2604–2612, 2022.
- Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- Wei, C., Wang, W., Yang, W., and Liu, J. Deep retinex decomposition for low-light enhancement. In *British Machine Vision Conference (BMVC)*, 2018.
- Wu, K., Lin, J., Li, Z., Miao, J., Xing, G., Luo, J., Zhao, H., Liu, Y., and Zhang, G. Device-independent low-light image enhancement via learnable von kries chromatic adaptation transform. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(9):13488–13503, 2026a.
- Wu, W., Weng, J., Zhang, P., Wang, X., Yang, W., and Jiang, J. URetinex-Net: Retinex-based deep unfolding network for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5891–5900, 2022.
- Wu, Y., Zhang, X., Chen, J., Chang, L., Ma, J., Chen, S., Zhu, L., Di, J., and Zhang, H. Thresholded cross-attention for reliable intensity-chromaticity fusion in low-light image enhancement. *arXiv preprint arXiv:2607.13925*, 2026b.
- Xu, X., Wang, R., Fu, C.-W., and Jia, J. SNR-aware low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 17693–17703, 2022.
- Yan, Q., Feng, Y., Zhang, C., Wang, P., Wu, P., Dong, W., Sun, J., and Zhang, Y. You only need one color space: An efficient network for low-light image enhancement. *arXiv preprint arXiv:2402.05809*, 2024.

- Yan, Q., Feng, Y., Zhang, C., Pang, G., Shi, K., Wu, P., Dong, W., Sun, J., and Zhang, Y. HVI: A new color space for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5678–5687, 2025.
- Yang, J., Yang, B., Ge, H., Liang, Y., Lee, H. P., and Wu, C. RHVI-FDD: A hierarchical decoupling framework for low-light image enhancement. *arXiv preprint arXiv:2604.05781*, 2026.
- Yang, W., Wang, W., Huang, H., Wang, S., and Liu, J. Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Transactions on Image Processing*, 30:2072–2086, 2021.
- Yi, X., Xu, H., Zhang, H., Tang, L., and Ma, J. Diff-retinex: Rethinking low-light image enhancement with a generative diffusion model. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 12268–12277, 2023.
- Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., and Yang, M.-H. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5718–5729, 2022.
- Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 586–595, 2018.
- Zhang, Y., Zhang, J., and Guo, X. Kindling the darkness: A practical low-light image enhancer. In *Proceedings of the 27th ACM International Conference on Multimedia (MM)*, pp. 1632–1640, 2019.
- Zhang, Z., Zheng, H., Hong, R., Xu, M., Yan, S., and Wang, M. Deep color consistent network for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1899–1908, 2022.
- Zhou, S., Li, C., and Loy, C. C. LEDNet: Joint low-light enhancement and deblurring in the dark. In *Computer Vision – ECCV 2022*, pp. 573–589. Springer, 2022.

A. Related Work

Retinex-based enhancement. Retinex theory (Land, 1977) models an image as the product of illumination and reflectance, and most learned low-light enhancers descend from it. RetinexNet (Wei et al., 2018) learns the decomposition end to end and introduced the LOL-v1 benchmark; KinD (Zhang et al., 2019) enhances illumination and reflectance along separate branches; URetinex-Net (Wu et al., 2022) and RUAS (Liu et al., 2021) unroll the decomposition into a network, the latter with architecture search. Retinexformer (Cai et al., 2023) folds the illumination estimate into a one-stage transformer and is the published model closest in size to HVI-CIDNet.

Supervised enhancers in other formulations. LLFlow (Wang et al., 2022) models the conditional distribution of normal-light images with a normalizing flow; SNR-Aware (Xu et al., 2022) routes between local and global processing by a spatial signal-to-noise estimate; LLFormer (Wang et al., 2023) targets ultra-high-definition inputs; GSAD (Hou et al., 2023) and Diff-Retinex (Yi et al., 2023) use diffusion models. The channel-attention block inside HVI-CIDNet follows the design of Restormer (Zamir et al., 2022). LEDNet (Zhou et al., 2022) treats enhancement and deblurring jointly and introduced the LOL-Blur dataset.

Unsupervised and zero-reference methods. Zero-DCE (Guo et al., 2020) estimates per-pixel enhancement curves without paired data; EnlightenGAN (Jiang et al., 2021) trains adversarially on unpaired images; SCI (Ma et al., 2022) learns a self-calibrated illumination map with a very small network. HVI-CIDNet is trained with pairs only and its authors list unsupervised training as open.

Color spaces for enhancement. Most of the methods above operate on sRGB, and the HSV space that HVI-CIDNet starts from dates to Smith (1978). Bread (Guo & Hu, 2023) splits an image into luminance and chrominance and lets the enhanced luminance guide a chrominance mapper; LYT-Net (Brateanu et al., 2025) processes Y and UV along separate lightweight paths; DCC-Net (Zhang et al., 2022) separates a gray image from an explicit color histogram; Ucolor (Li et al., 2021) embeds RGB, HSV and Lab features jointly; CoLIE (Chobola et al., 2024) fits an implicit illumination field in HSV; CSTNet (Guan et al., 2025) learns a YCbCr-like transform for nighttime deraining, and CATformer (Wu et al., 2026a) learns a von Kries adaptation before CIELAB. HVI-CIDNet (Yan et al., 2025) instead maps sRGB into a polar Horizontal/Vertical/Intensity (HVI) space in which hue is an angle, saturation a radius, and the radius is shrunk toward the origin as intensity goes to zero by a learned collapse function; the network enhances intensity and chroma along two interacting branches and is supervised in both sRGB and HVI. An earlier version of the method (Yan et al., 2024) and a successor for extreme darkness (Shi et al., 2026b) exist; we study the CVPR 2025 version whose code is public.

Work that builds on or questions HVI. FusionNet (Shi et al., 2025) fuses HVI-CIDNet with two sRGB enhancers and won the NTIRE 2025 challenge (Liu et al., 2025); DiffLight (Feng et al., 2024) and Q-CIDNet (Verma et al., 2025) reuse the network for diffusion-based and video enhancement, and JarvisIR (Lin et al., 2025) calls it as an off-the-shelf tool. Several works modify the intensity or the fusion: RHVI-FDD (Yang et al., 2026) argues that the max-RGB intensity is sensitive to noise spikes, BC-IHV (Ai et al., 2026) replaces the linear intensity by a learnable Box-Cox law, VCR (Cheng et al., 2026) recalibrates channels by their variance, HAIMNet (Wang et al., 2026) and TCA-Net (Wu et al., 2026b) change how the two branches interact, and TPCNet (Shi et al., 2026a) adds physical constraints. Others report sensitivity of HVI-CIDNet to input brightness (Pilligua et al., 2025; Du et al., 2026) or to per-pair photometric mismatch in the training data (Li et al., 2026). None of these reports a seed spread, and none compares color spaces under one fixed network with more than one run.

Evaluation practice. Model selection on the test split is a documented source of inflated benchmark numbers (Musgrave et al., 2020; Recht et al., 2019), and single-run comparisons hide variance that changes conclusions (Bouthillier et al., 2021; Pineau et al., 2021). For low-light enhancement in particular, Nguyen et al. (2024) document scene overlap between the LOL-v1 train and test splits, and Liao et al. (2025) show that the brightness mismatch hidden by GT-mean rescaling is systematic. We therefore select checkpoints on a scene-disjoint validation split, read the test split once, report every mean with its seed spread, and report the GT-mean and gated variants separately from the primary numbers.

Datasets and metrics. We use the paired LOL-v1 benchmark of Wei et al. (2018); its successor LOL-v2 (Yang et al., 2021), SID (Chen et al., 2018), SICE (Cai et al., 2018), and LOL-Blur (Zhou et al., 2022) are the other paired sets on which HVI-CIDNet was reported. Full-reference quality is measured by PSNR, SSIM (Wang et al., 2004), and LPIPS

(Zhang et al., 2018); no-reference quality by NIQE (Mittal et al., 2013) and BRISQUE (Mittal et al., 2012). The perceptual loss term of HVI-CIDNet is the VGG feature loss of Johnson et al. (2016). Color fidelity, when we report it, uses the CIEDE2000 difference (Sharma et al., 2005).

B. Additional Experiments

Full tables, per-image analysis, and protocol details are reported here once the studies have run.

B.1. Protocol Details

Each run directory records the training-script SHA-256, the upstream HVI-CIDNet commit, the split-file SHA-256, the torch version, and the GPU name. The validation summary written at the end of every run is the only source of the numbers in this paper.